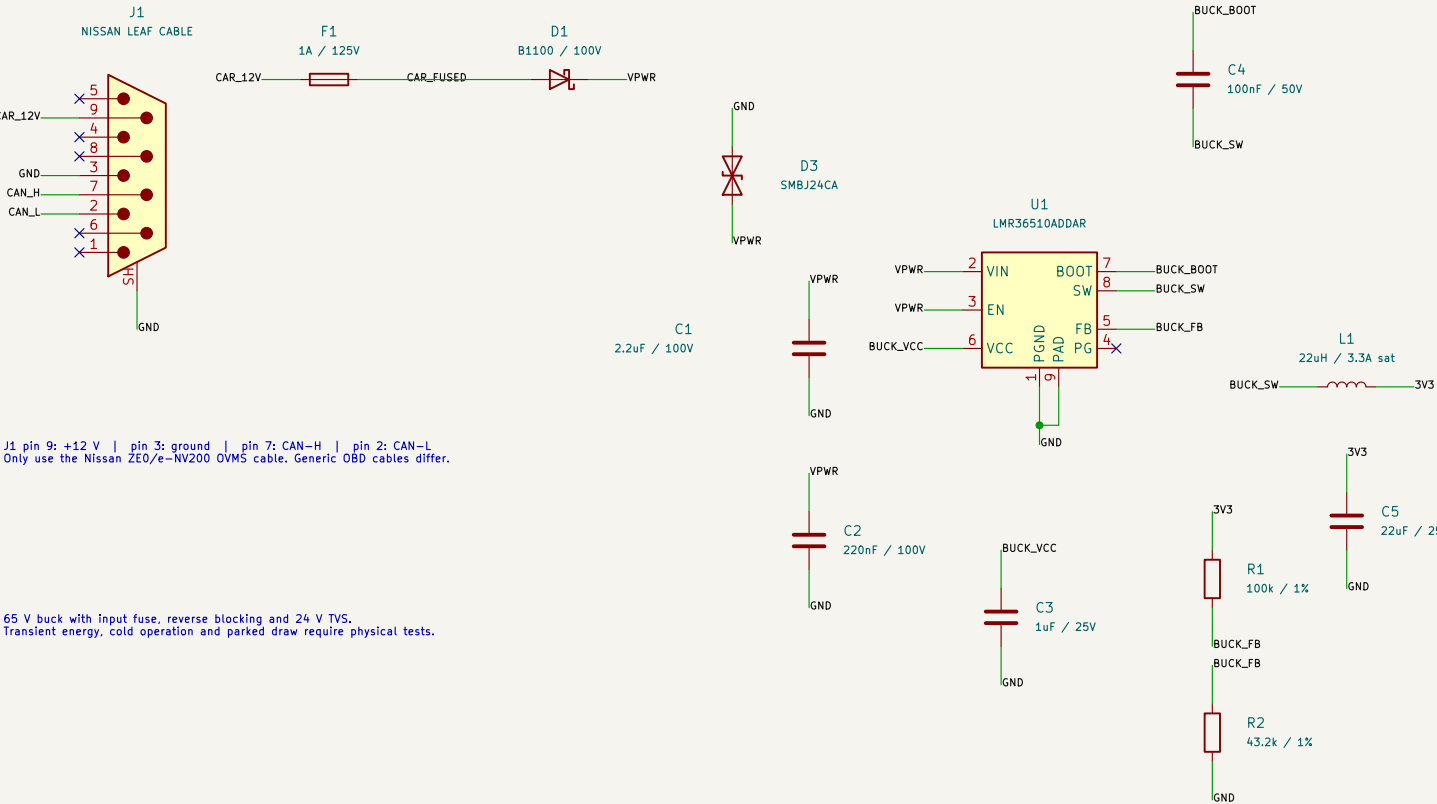


LEAF HEAT / Wi-Fi + one CAN channel

REV E – HAND–SOLDER PROTOTYPE – FACTORY: U1 / U2 / U4 / L1

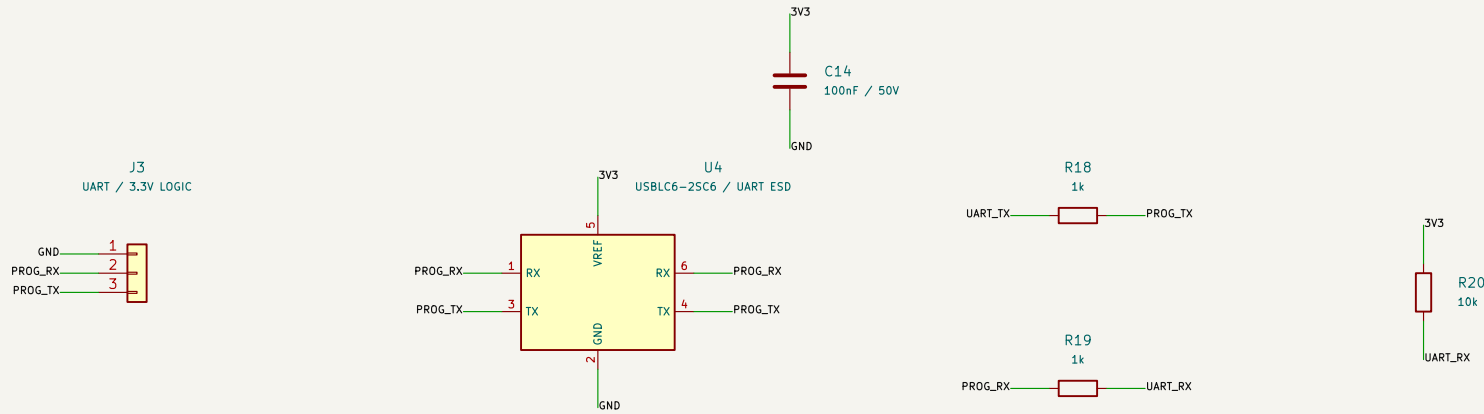
1 VEHICLE POWER / 3.3 V SUPPLY



J1 pin 9: +12 V | pin 3: ground | pin 7: CAN–H | pin 2: CAN–L
Only use the Nissan ZEO/e–NV200 OVMS cable. Generic OBD cables differ.

65 V buck with input fuse, reverse blocking and 24 V TVS.
Transient energy, cold operation and parked draw require physical tests.

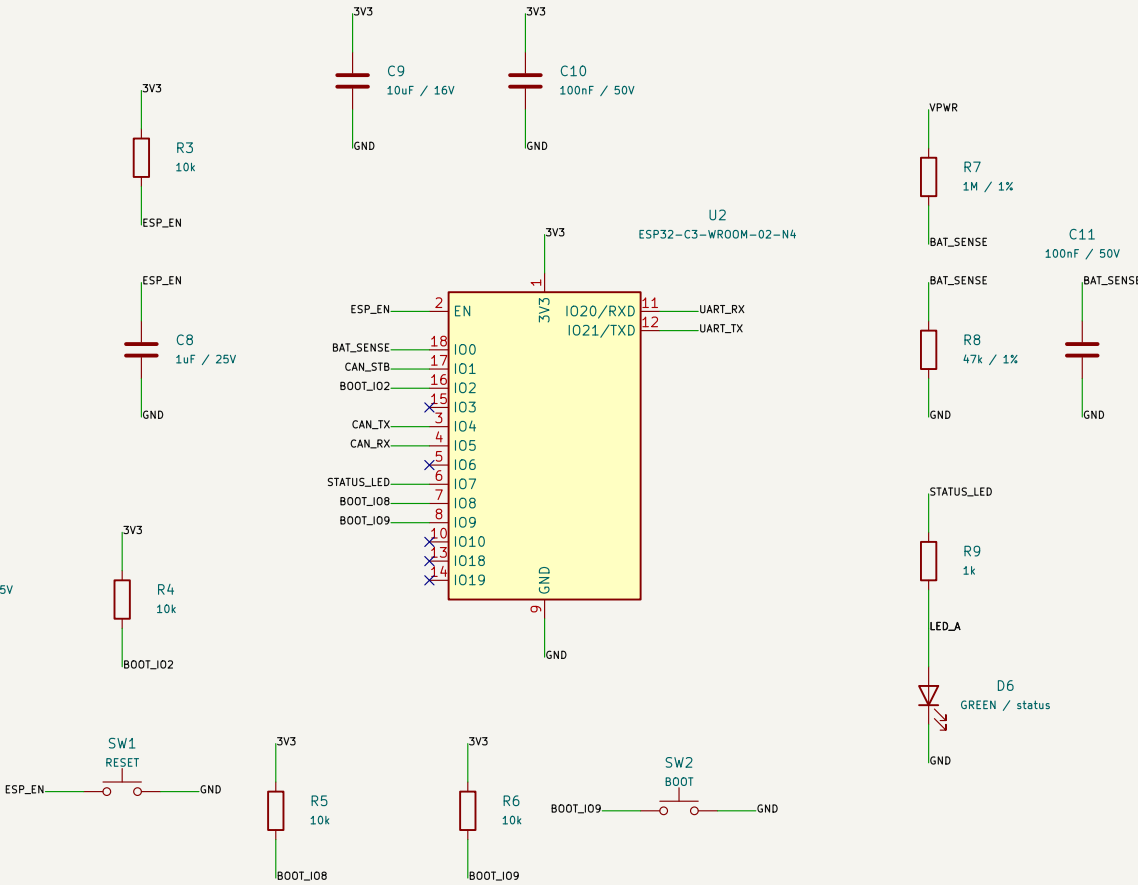
3 UART PROGRAMMING – NO USB / NO PROGRAMMER POWER



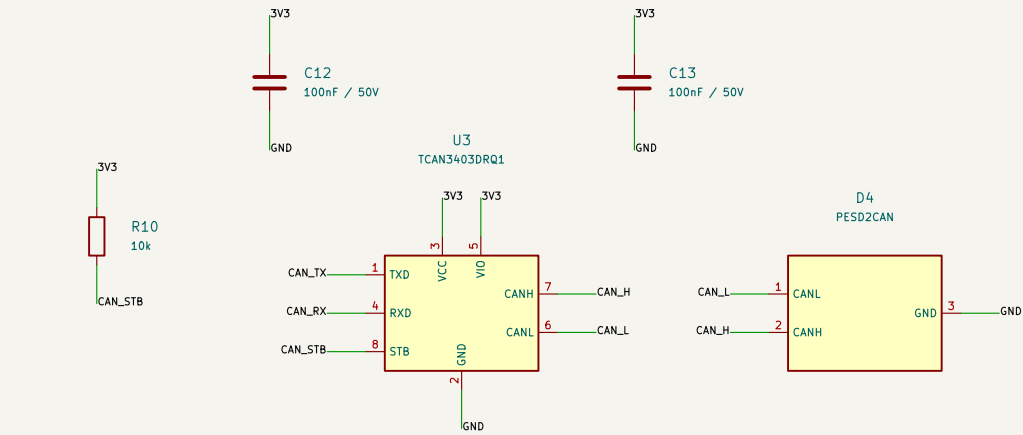
J3: 1 GND | 2 board RX | 3 board TX, 3.3 V logic only.
Power from a current–(limited 12 V bench supply via J1 (9 +, 3 GND)).
Vehicle unplugged, Adapter power wire unused. Connect only with board powered.
Hold BOOT: press/release RESET; release BOOT. Flash via UART0 at 115200 baud.

GPIO3 / GPIO18 / GPIO19 unused. No USB–presence sensing in Rev E.
OTA updates need firmware implementation. UART remains the recovery interface.

2 Wi-Fi / BOOT / BATTERY SENSE



4 VEHICLE CAN – NO TERMINATION



CAN TX: GPIO4 | RX: GPIO5 | STB: GPIO1
STB is pulled high during reset. Drive low only for a requested action.
500 kbit/s, standard 11–bit frames; 2013–2015 Leaf profile to be confirmed.

Voltage sense is after the input diode; calibration required.
Software low–battery sleep is required; there is no hardware battery disconnect.

Engineering review and vehicle tests required before release

DIY Leaf project

Sheet: /
File: leaf–heat–v5.kicad_sch

Title: Leaf Heat – Wi-Fi climate control

Size: A3 Date: 2026–09–12

KiCad E.D.A. 10.0.6

Rev: E

Id: 1/1